

Productivity, Output, and Employment

By Instructors : Macroeconomics-I, 2025-26

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Learning Objectives

1. Discuss production function properties and changes (Section 3.1)
2. Discuss factors that affect the demand for labor (Section 3.2)
3. Discuss the factors that affect the supply of labor (Section 3.3)
4. Identify factors that affect labor market equilibrium (Section 3.4)
5. Explain how the unemployment rate is measured and describe changes in employment status (Section 3.5)
6. Explain the significance of Okun's law (Section 3.6)

How Much Does the Economy Produce? The Production Function

Factors of Production

The economy relies on several key factors of production:

Key Factors of Production:

- Capital (K)
- Labor (N)
- Others (raw materials, land, energy)
- Productivity of factors depends on technology and management

The Production Function

The production function describes the relationship between inputs and output:

$$Y = AF(K, N) \quad (1)$$

where parameter A is **total factor productivity**, representing the effectiveness with which capital and labor are used.

- **Constant Returns to Scale:** When $\alpha + \beta = 1$, the function exhibits constant returns to scale, meaning doubling all inputs doubles output. This became a cornerstone assumption in neoclassical economics.
- The parameters α and β or $(1 - \alpha)$ in the Cobb-Douglas production function correspond to the **production factor elasticity of labor**

Application: The Cobb-Douglas production function works well for the U.S. economy: $Y = AK^{0.3}N^{0.7}$. This functional form has been extensively validated using U.S. economic data.

Also note The production function applies both to an economy as a whole (where Y , K , and N refer to the economy's output, capital stock, and number of workers, respectively) and to an individual firm, in which case Y , K , and N refer to the firm's output, capital, and number of workers, respectively.

- **Mathematical proof for constant returns to scale:** If $Y = AK^\alpha N^\beta$ and we multiply all inputs by factor λ : $Y' = A(\lambda K)^\alpha (\lambda N)^\beta = A\lambda^{\alpha+\beta} K^\alpha N^\beta$. When $\alpha + \beta = 1$, we get $Y' = \lambda Y$, so doubling inputs ($\lambda = 2$) exactly doubles output.
- **Production elasticity derivation:**
Capital elasticity = $\frac{\partial \ln Y}{\partial \ln K} = \frac{\partial Y / Y}{\partial K / K}$.
For $Y = AK^\alpha N^\beta$: $\frac{\partial Y}{\partial K} = \alpha AK^{\alpha-1} N^\beta$,
so elasticity = $\frac{\alpha AK^{\alpha-1} N^\beta \cdot K}{AK^\alpha N^\beta} = \alpha$.
Similarly for labor: $\frac{\partial \ln Y}{\partial \ln N} = \beta$.

and capital, respectively. Observing the actual shares of income received by capital and labor provides a way of estimating the parameter α

The Shape of the Production Function

Production functions exhibit two main properties:

1. **Slopes upward:** More of any input produces more output
2. **Slope becomes flatter as input rises:** Diminishing marginal product as input increases

Marginal Product of Capital (MPK):

$$MPK = \Delta Y / \Delta K \quad (2)$$

Properties:

- Equal to slope of production function graph (Y vs. K)
- MPK always positive
- Diminishing marginal productivity — MPK declines as K rises

Marginal Product of Labor (MPN):

$$MPN = \Delta Y / \Delta N \quad (3)$$

Properties:

- Equal to slope of production function graph (Y vs. N)
- MPN always positive
- Diminishing marginal productivity of labor

The easiest way to graph it is to hold both total factor productivity and one of the two factors of production, either capital or labor, constant and then graph the relationship between output and the other factor.

Supply Shocks

Supply Shock Definition:

A supply shock (also called a productivity shock) is a change in an economy's production function. Supply shocks affect the amount of output that can be produced for a given amount of inputs.

The Dual Interpretation

- In competitive markets, factors are paid their **marginal products**
 - Wage rate = Marginal product of labor (MPN)
 - Rental rate = Marginal product of capital (MPK)
 - Marginal products for $Y = AK^\alpha N^\beta$: $MPK = \frac{\partial Y}{\partial K} = \alpha AK^{\alpha-1} N^\beta$ and $MPN = \frac{\partial Y}{\partial N} = \beta AK^\alpha N^{\beta-1}$.
- **Income Shares:**
 - Labor's share of total income = β
 - Capital's share of total income = α
 - Total shares: $\alpha + \beta = 1$
 - Income share calculation:
 Labor's share = $\frac{w \cdot N}{Y} = \frac{MPN \cdot N}{Y} = \frac{\beta AK^\alpha N^{\beta-1} \cdot N}{Y} = \frac{\beta AK^\alpha N^\beta}{Y} = \beta \frac{Y}{Y} = \beta$.
 Similarly, Capital's share = $\frac{r \cdot K}{Y} = \frac{MPK \cdot K}{Y} = \frac{\alpha AK^{\alpha-1} N^\beta \cdot K}{Y} = \frac{\alpha AK^\alpha N^\beta}{Y} = \alpha$.
- This duality between production elasticities and income shares is unique to the Cobb-Douglas function and assumes competitive markets. It's why this functional form became so popular in economics - it elegantly connects technology (production function) with distribution (income shares).

Shocks may be:

- **Positive:** Increasing output (e.g., technological innovations)
 - **Negative (Adverse):** Decreasing output (e.g., oil price increases)
- Examples of supply shocks:** weather events, inventions and innovations, government regulations, oil prices

Supply shocks shift the graph of the production function:

- **Negative shock:** Usually slope of production function decreases at each level of input (parameter A declines)
- **Positive shock:** Usually slope of production function increases at each level of output (parameter A increases)

The Demand for Labor

How Much Labor Do Firms Want to Use?

We analyze labor demand under the following assumptions:

1. Hold capital stock fixed – short-run analysis
2. Workers are all alike
3. Labor market is competitive
4. Firms maximize profits

Analysis at the Margin

Firms analyze the costs and benefits of hiring one extra worker:

Profit-Maximizing Condition:

Firms' profits are highest when:

$$w = MPN \quad (4)$$

where w is the real wage.

- If $w > MPN$: Firm pays marginal worker more than worker produces – reduce workers
- If $w < MPN$: Marginal worker produces more than paid – hire more workers

There are two ways to graph labor demand that differ importantly.

- The first plots labor (N) against the nominal wage (W) based on $W = P \times MPL$, holding price level P constant. This is downward sloping but shifts when P changes because the firm needs higher nominal wages to hire the same labor when prices rise.
- The second is the true labor demand curve, plotting N against the real wage (W/P) based on $MPL = W/P$. This curve also slopes downward but does not shift when P changes, because if W and P both rise proportionally, the real wage stays constant and labor demand is unchanged.
- The key difference is that the N vs. W graph is conditional on a fixed P and shifts with price changes, while the labor demand curve (N vs. W/P) is a structural relationship based on technology that remains stable when prices change. Firms ultimately care about real costs (W/P), not nominal wages, making the real wage relationship the true labor demand curve.

The Marginal Product of Labor and Labor Demand: An Example

Consider a firm setting the nominal wage equal to the marginal revenue product of labor:

$$MRPN = P \times MPN \quad (5)$$

The condition $W = MRPN$ is equivalent to $w = MPN$, since $W = P \times w$ and $MRPN = P \times MPN$.

When the wage changes

- **Wage rises:** $W > MRPN$ unless N is reduced so $MRPN$ rises
- **Wage falls:** $W < MRPN$ unless N rises so $MRPN$ falls

The Labor Demand Curve

The **labor demand curve** shows the relationship between the real wage rate and the quantity of labor demanded.

Key properties:

- It is the same as the MPN curve (since $w = MPN$ at equilibrium)
- Downward sloping: Higher real wage $\hat{w}$ $\rightarrow$ less labor demanded

Theoretical Application: The field of labor economics studies determinants of labor supply. A major issue in the 1970s and early 1980s concerned increased participation rates of women. Research explained why there is a negative relationship between family income and labor force participation across families, but a positive relationship over time. In the 2010s, research focused on explaining the significant decline in labor force participation, especially since the 2007–2009 recession.

Factors That Shift the Labor Demand Curve

Important: A change in the wage causes a movement along the labor demand curve, not a shift of the curve.

Factors that shift the labor demand curve:

1. **Supply shocks:** Beneficial supply shock raises MPN $\hat{w}$ $\rightarrow$ shifts labor demand curve right; adverse shock shifts it left
2. **Size of capital stock:** Higher capital stock raises MPN $\hat{w}$ $\rightarrow$ shifts labor demand curve right; lower capital stock shifts it left

Aggregate Labor Demand

Aggregate labor demand is the sum of all firms' labor demand. The same factors (supply shocks, size of capital stock) that shift individual firms' labor demand cause shifts in aggregate labor demand.

The Supply of Labor

The Income-Leisure Trade-off

Individual labor supply depends on the labor-leisure choice:

- Utility depends on both consumption and leisure
- Must compare costs and benefits of working another day
 - **Costs:** Loss of leisure time
 - **Benefits:** More consumption (higher income)
- Work until benefits equal costs

Real Wages and Labor Supply

An increase in the real wage has offsetting income and substitution effects:

Labor supply can increase in two ways:

- **Intensive margin:** Changes in hours worked of existing workers
- **Extensive margin:** Changes in number of workers hired
- However, as Keane and Rogerson point out, changes in aggregate hours worked over both short and long periods of time are influenced in an important way by choices at the extensive margin; that is, the choices of individuals about whether to work or not.
- In fact, if we take into account the extensive margin, the aggregate labor supply elasticity is much higher.

Substitution Effect: Higher real wage encourages work, since the reward for working is higher

Income Effect: Higher real wage increases income for the same work time; with higher income, can afford more leisure
 → supply less labor

A Pure Substitution Effect: One-Day Rise in Real Wage

A temporary real wage increase has just a pure substitution effect, since the effect on wealth is negligible.

A Pure Income Effect: Winning the Lottery

Winning the lottery doesn't have a substitution effect (doesn't affect reward for working), but makes person wealthier → consume more goods and take more leisure.

Combined Effects: Long-Term Increase in Real Wage

- Reward to working is greater: substitution effect toward more work
- With higher wage, don't need to work as much: income effect toward less work
- The longer the high wage lasts, the stronger the income effect

Empirical Evidence:

- Labor supply increases with temporary rise in real wage
- Labor supply falls with permanent increase in real wage

- Thus, in our macroeconomic model, it is most useful to think of the representative consumer as a fictitious person who stands in for the average consumer in the economy.
- Hours of work for this fictitious person should be interpreted as average hours of work in the whole economy.
- Thus, when aggregate hours of work change in practice, because of changes along the intensive and extensive margins, we should think of this as changes in the representative consumer's hours of work in our model.
- Also there are many different wages in the economy; our one-wage model is a simplification.

The Labor Supply Curve

The labor supply curve relates quantity of labor supplied to real wage. It slopes upward because higher wage encourages people to work more.

Factors That Shift the Labor Supply Curve

1. **Wealth:** Higher wealth reduces labor supply (shifts curve left)
2. **Expected future real wage:** Higher expected future wage is like increase in wealth reduces labor supply (shifts curve left)

Aggregate Labor Supply

Aggregate labor supply rises when current real wage rises:

- Some people work more hours
- Other people enter labor force
- Result: Aggregate labor supply curve slopes upward

Factors increasing labor supply:

- Decrease in wealth
- Decrease in expected future real wage

- Increase in working-age population (higher birth rate, immigration)
- Increase in labor force participation (increased female participation, elimination of mandatory retirement)

Labor Market Equilibrium

Classical Model of the Labor Market

Labor Market Equilibrium:

Labor supply equals labor demand. The classical model assumes the real wage adjusts quickly.

Equilibrium determines:

- Full-employment level of employment ($\bar{N}$)
- Market-clearing real wage ($\bar{w}$)

Factors that shift labor supply or demand affect $\bar{N}$ and $\bar{w}$.

Limitation: The classical model can't study unemployment (examined in Chapter 11 with models where real wage doesn't adjust quickly).

- Assumes all unemployment is voluntary (people choosing not to work at prevailing wages)
- Cannot explain persistent involuntary unemployment observed in real economies
- No role for structural, frictional, or cyclical unemployment beyond "natural rate"
- Doesn't account for job search frictions, matching problems, or skill mismatches

Full-Employment Output

Categories: employed, unemployed, not in the labor force.

Full-Employment Output (also called **potential output**, $\bar{Y}$) is the level of output when the labor market is in equilibrium:

$$\bar{Y} = AF(K, \bar{N}) \quad (6)$$

$\bar{Y}$ is affected by changes in $\bar{N}$ or the production function (e.g., supply shocks).

Instantaneous Market Clearance Assumption:

- **Wage rigidity ignored:** Wages don't adjust instantly due to contracts, minimum wages, efficiency wages, union bargaining
- **Price stickiness ignored:** Many prices are set for periods and don't change immediately
- **Coordination failures:** Markets may not clear simultaneously across all sectors
- **Information problems:** Perfect information assumption unrealistic

Unemployment

Why Do We Care About Unemployment?

- High or low unemployment is a symptom of inefficiency
- High unemployment: many resources are idle
- Low unemployment: labor shortages $\Rightarrow$ so many resources are wasted trying to recruit workers
- Unemployment reduces the welfare of the unemployed, especially those unemployed for long periods of time
- Low mental and physical health (anxiety, low self esteem, low life satisfaction), even after controlling for income

Changes in Employment Status:

Flows occur between categories: employed, unemployed, and not in labor force.

Discouraged workers: People who have become so discouraged by lack of job-finding success that they stop searching.

Measuring Unemployment

Key Definitions:

$$\text{Labor Force} = \text{Employed} + \text{Unemployed} \quad (7)$$

$$\text{Unemployment Rate} = \frac{\text{Unemployed}}{\text{Labor Force}} \quad (8)$$

$$\text{Participation Rate} = \frac{\text{Labor Force}}{\text{Adult Population}} \quad (9)$$

$$\text{Employment Ratio} = \frac{\text{Employed}}{\text{Adult Population}} \quad (10)$$

How Long Are People Unemployed?

Key Facts About Unemployment Duration:

1. Most unemployment spells are short duration
 - Unemployment spell = period of continuous unemployment
 - Duration = length of unemployment spell
2. Most unemployed people on a given date are experiencing long-duration spells

Result:

- 28 unemployment spells during year: 24 short (one month), 4 long (one year) $\hat{=}$ most spells are short
- At any date, unemployment = 6: four have long spells (one year), two have short spells (one month) $\hat{=}$ most unemployed people have long spells

Numerical Example Reconciling These Facts:

Consider a labor force of 100. On the first day of every month, two workers become unemployed for one month each. On the first day of every year, four workers become unemployed for one year each.

Policy Application: Because labor market matching takes time, government policy often provides income support via unemployment insurance. But this has disincentive effects—people may prolong job searches to receive benefits longer, getting income without working. Some economists believe high average European unemployment rates in the 1980s partly resulted from generous unemployment insurance.

Why There Are Always Unemployed People

Frictional Unemployment Search activity of firms and workers due to heterogeneity. The matching process takes time.

Structural Unemployment Long-term and chronic unemployment existing even when the economy is not in recession.

Causes:

- Lack of skills prevents some workers from finding long-term employment
- Reallocation of workers out of shrinking industries or depressed regions; matching takes long time

Natural Rate of Unemployment ($\bar{u}$):

The unemployment rate when output and employment are at full-employment levels.

$$\bar{u} = \text{Frictional} + \text{Structural Unemployment} \quad (11)$$

Cyclical unemployment: Difference between actual unemployment rate and natural rate: $(u - \bar{u})$

Data Application: Much controversy exists about measuring the natural rate of unemployment (discussed in greater detail in Chapter 12). The problem: natural rate isn't directly observable, so we must use available data to estimate it based on some labor market model. This is quite difficult, and much debate exists among empirical macroeconomists about the natural rate's size at any date.

Data Application: What is more closely related to output: Employment growth (from establishment survey) or unemployment rate? Business economists usually focus on employment growth (or average hours worked) rather than unemployment rate, partly because labor force participation rate changes lead to unemployment rate fluctuations not correlated with output.

Alternative Measures of the Unemployment Rate

1. **U-1:** unemployed 15 weeks or more
2. **U-2:** counts job losers or persons who completed temporary jobs, so it does not count people who have quit their jobs
3. **U-3:** the official rate, as defined in textbook
4. **U-4:** like U-3 but adds discouraged workers (those not looking for work because they do not think they can find a job)
5. **U-5:** like U-4 but adds marginally attached workers (those who say they are not looking for work but indicate that they want and are available for a job and have looked for work sometime in the past 12 months, Have NOT looked in the **past 4 weeks**)
6. **U-6:** like U-5 but adds persons who are employed part time for economic reasons (those who want and are available for full-time work but have had to settle for a part-time schedule)

Relating Output and Unemployment: Okun's Law

Okun's Law: Relationship between output (relative to full-employment output) and cyclical unemployment:

$$\frac{\bar{Y} - Y}{\bar{Y}} = 2(u - \bar{u}) \quad (12)$$

Alternative formulation (if average growth rate of full-employment output is 3%):

$$\Delta Y / Y = 3 - 2\Delta u \quad (13)$$

Result: 2% reduction in output associated with 1 percentage point increase in unemployment rate.

Why is the Okun's Law Coefficient 2, Not 1?

When cyclical unemployment rises, other things happen:

- Labor force falls, Hours of work per worker decline
- Average productivity of labor declines

Key Equations Summary

Key Terms

- | | |
|---------------------------------------|-----------------------------------|
| • aggregate demand for labor | (MPK) |
| • aggregate supply of labor | • marginal product of labor |
| • chronically unemployed | (MPN) |
| • cyclical unemployment | • marginal revenue product of |
| • diminishing marginal productivity | labor (MRPN) |
| • discouraged workers | • natural rate of unemployment |
| • duration | • Okun's law |
| • employment ratio | • participation rate |
| • factors of production | • production function |
| • frictional unemployment | • productivity |
| • full-employment level of employment | • real wage |
| • full-employment output | • structural unemployment |
| • income effect of a higher real wage | • substitution effect of a higher |
| • labor force | real wage |
| • leisure | • supply shock |
| • marginal product of capital | • total factor productivity |
| | • unemployment rate |
| | • unemployment spell |

Equation

$$Y = AF(K, N)$$

$$Y = AK^{0.3}N^{0.7}$$

$$MPN = w$$

$$MRPN = P \times MPN$$

$$\bar{Y} = AF(K, \bar{N})$$

$$(Y - \bar{Y}) / \bar{Y} = 2(u - \bar{u})$$

$$\Delta Y / Y = 3 - 2\Delta u$$

Description

Production function

Cobb-Douglas PF

Marginal product of Labour

Marginal revenue product

Full-employment output

Okun's law (gap version)

Okun's law (growth version)

Appendix (Additional Notes for Curious Minds Only)

The Cobb-Douglas Production Function: Mathematical Note

In general, a production function is a specification of how the quantity of output behaves as a function of the inputs used in production. This concept can be applied at the level of individual firms, industries, or entire economies. Since we're doing macroeconomics we will be considering an aggregate production function, applying at the economy-wide level.

Various specific mathematical forms have been put forward for the production function, but the most commonly used is that developed by Charles Cobb and Paul Douglas in the second quarter of the 20th century. Here's their specification:

The Cobb-Douglas Production Function

$$Y = AK^\alpha N^{1-\alpha} \quad 0 < \alpha < 1 \quad (14)$$

Where:

- Y = aggregate output
- K = capital input
- N = labor input
- A = Total Factor Productivity (TFP)
- α = capital's share parameter

The A term represents Total Factor Productivity (TFP for short); you can think of this as a "quality" factor—as opposed to K and N which are just quantitative. The value of A reflects the state of technology as well as the skill and education level of the workforce. All being well, we'd expect A to be gradually increasing over time.

Marginal Product, Diminishing Returns

A particularly important aspect of a production function is the marginal product of the factors. Take first the marginal product of labor (or MPN for short)—that is, the change in output that results when the labor input is varied, holding the capital input and TFP constant. We find this by taking the first derivative of equation (1) with respect to N :

Courtesy: Allin Cottrell

A production function shows the relationship between inputs (factors of production) and outputs.

Developed by mathematician Charles Cobb and economist Paul Douglas in the 1920s-1930s.

TFP captures technological progress, education, institutions, and other factors affecting productivity.

Marginal Product of Labor (MPN): the additional output from one more unit of labor, holding capital constant.

$$\text{MPN} = \frac{dY}{dN} = (1 - \alpha)AK^\alpha N^{1-\alpha-1} \quad (15)$$

$$= (1 - \alpha) \frac{AK^\alpha N^{1-\alpha}}{N^1} \quad (16)$$

$$= (1 - \alpha) \frac{Y}{N} > 0 \quad (17)$$

Given that Y and N must be positive and α is a positive fraction, we see that the marginal product of labor must be positive: a greater labor input leads to the production of more output. No surprise there. The familiar economic concept of “diminishing returns” leads us to expect that the MPN, while positive, should be declining: as the labor input is increased, holding K and TFP constant, output should increase but at a diminishing rate. Does the Cobb-Douglas function satisfy this condition? To find out we need to take the derivative of the MPN with respect to N , or in other words the second derivative of Y with respect to N .

Diminishing returns: each additional unit of input adds less output than the previous unit.

$$\frac{d\text{MPN}}{dN} = \frac{d^2Y}{dN^2} = (-\alpha)(1 - \alpha)AK^\alpha N^{1-\alpha-2} \quad (18)$$

$$= (-\alpha)(1 - \alpha) \frac{AK^\alpha N^{1-\alpha}}{N^2} \quad (19)$$

$$= (-\alpha)(1 - \alpha) \frac{Y}{N^2} < 0 \quad (20)$$

We can tell that the second derivative is negative—hence satisfying diminishing returns—because all terms in the multiplicative expression are positive apart from the negative $-\alpha$.

Strictly analogous math tells us that the Cobb-Douglas function also exhibits a positive but diminishing marginal product of capital, MPK. (In this case the thought-experiment is, what happens to output when K is increased while N and TFP are held constant?)

The same logic applies to capital: positive but diminishing marginal returns.

Marginal Products in Cobb-Douglas

Marginal Product of Labor:

$$\text{MPN} = (1 - \alpha) \frac{Y}{N} > 0$$

Marginal Product of Capital:

$$\text{MPK} = \alpha \frac{Y}{K} > 0$$

Both marginal products are positive but diminishing.

Diminishing returns to capital:

$$\frac{dMPK}{dK} = \frac{d^2Y}{dK^2} = (\alpha - 1)\alpha AK^{\alpha-2}N^{1-\alpha} = (\alpha - 1)\alpha \frac{Y}{K^2} < 0 \quad (21)$$

Cross Partial

A further point relevant for macroeconomic analysis: what (if anything) happens to the marginal product of labor when the capital input is increased? And conversely, what happens to the MPK when N increases? In mathematical terms, we're talking about the so-called "cross-partial" derivatives, $dMPN/dK$ and $dMPK/dN$.

Cross-partial derivatives measure how one factor's marginal product changes when the other factor increases.

$$\frac{dMPN}{dK} = \alpha(1 - \alpha)AK^{\alpha-1}N^{1-\alpha-1} \quad (22)$$

$$= \alpha(1 - \alpha)\frac{AK^{\alpha}N^{1-\alpha}}{K^1N^1} \quad (23)$$

$$= \alpha(1 - \alpha)\frac{Y}{KN} > 0 \quad (24)$$

So an increase in capital raises the marginal product of labor. And

More capital makes labor more productive. This is called capital-labor complementarity.

$$\frac{dMPK}{dN} = (1 - \alpha)\alpha AK^{\alpha-1}N^{1-\alpha-1} \quad (25)$$

$$= (1 - \alpha)\alpha \frac{Y}{KN} > 0 \quad (26)$$

So raising N also raises the MPK. (And it turns out that the two cross partials are identical.)

Also note: from equations (2a) and (3a) it should be clear that an increase in Total Factor Productivity, A , will raise the marginal products of both factors.

Technological progress (higher A) increases the productivity of both labor and capital.

Returns to Scale

We've shown that the Cobb-Douglas function gives diminishing returns to both labor and capital when each factor is varied in isolation. But what happens if we change both K and N in the same proportion?

Suppose an economy in an initial state has inputs K_0 and N_0 and produces output Y_0 :

$$Y_0 = AK_0^{\alpha}N_0^{1-\alpha} \quad (27)$$

Now suppose we scale the inputs by some common factor λ . (For example, $\lambda = 2$ would mean that we double each input.) We'll then have inputs $K_1 = \lambda K_0$ and $N_1 = \lambda N_0$ and will produce output Y_1 . The question is, how does Y_1 relate to Y_0 ? Let's see:

Returns to scale: what happens when we scale all inputs proportionally?

$$Y_1 = AK_1^\alpha N_1^{1-\alpha} \quad (28)$$

$$= A(\lambda K_0)^\alpha (\lambda N_0)^{1-\alpha} \quad (29)$$

$$= A\lambda^\alpha K_0^\alpha \lambda^{1-\alpha} N_0^{1-\alpha} \quad (30)$$

$$= \lambda^{\alpha+1-\alpha} AK_0^\alpha N_0^{1-\alpha} \quad (31)$$

$$= \lambda Y_0 \quad (32)$$

So if we scale both inputs by a common factor, the effect is to scale the output by that same factor. This is the defining characteristic of **constant returns to scale**. From the math above we can see that this occurs in the Cobb-Douglas function because the exponents on capital and labor, α and $1 - \alpha$, add up to 1.

Constant returns to scale: doubling all inputs doubles output.

General Case: Exponents and Returns to Scale

For a generalized production function $Y = AK^\alpha N^\beta$:

- If $\alpha + \beta > 1 \rightarrow$ **Increasing returns to scale**
- If $\alpha + \beta = 1 \rightarrow$ **Constant returns to scale**
- If $\alpha + \beta < 1 \rightarrow$ **Decreasing returns to scale**

The standard Cobb-Douglas has $\alpha + (1 - \alpha) = 1$, giving constant returns to scale.

Each exponent must be a positive fraction to ensure positive but diminishing marginal products.

Introduction to Returns to Scale

The term “returns to scale” refers to how well a business or company is producing its products. It tries to pinpoint increased production in relation to factors that contribute to production over a period of time. Most production functions include both labor and capital as factors. How can you tell if a function is increasing returns to scale, decreasing returns to scale, or having no effect on returns to scale? The three definitions below explain what happens when you increase all production inputs by a multiplier.

Returns to scale measures the responsiveness of output to a proportional change in all inputs.

The two primary factors of production are:

K = Capital (machinery, equipment, buildings)

L = Labor (workers, hours)

Multipliers

For illustrative purposes, we'll call the multiplier m . Suppose our inputs are capital and labor, and we double each of these ($m = 2$). We want to know if our output will more than double, less than double, or exactly double. This leads to the following definitions:

The multiplier m represents the factor by which we scale all inputs simultaneously.

Three Types of Returns to Scale:

- **Increasing Returns to Scale:** When our inputs are increased by m , our output increases by more than m .
- **Constant Returns to Scale:** When our inputs are increased by m , our output increases by exactly m .
- **Decreasing Returns to Scale:** When our inputs are increased by m , our output increases by less than m .

The multiplier must always be positive and greater than one because our goal is to look at what happens when we increase production. An m of 1.1 indicates that we've increased our inputs by 0.10 or 10 percent. An m of 3 indicates that we've tripled the inputs.

Mathematically: $m > 1$

Examples:

$m = 1.1$ means 10% increase

$m = 2$ means doubling inputs

$m = 3$ means tripling inputs

Three Examples of Economic Scale

Now let's look at a few production functions and see if we have increasing, decreasing, or constant returns to scale. Some textbooks use Q for quantity in the production function, and others use Y for output. These differences don't change the analysis, so use whichever your professor requires.

Note: Some textbooks use Q for quantity, others use Y for output. The analysis is identical.

Example 1: Linear Production Function

Example 1: $Q = 2K + 3L$

To determine the returns to scale, we will begin by increasing both K and L by m . Then we will create a new production function Q' . We will compare Q' to Q .

$$Q' = 2(Km) + 3(Lm) \quad (33)$$

$$= 2Km + 3Lm \quad (34)$$

$$= m(2K + 3L) \quad (35)$$

$$= mQ \quad (36)$$

After factoring, we can replace $(2K + 3L)$ with Q , as we were given that from the start. Since $Q' = mQ$ we note that by increasing all of our inputs by the multiplier m we've increased production by exactly m . As a result, we have **constant returns to scale**.

Linear production functions like $Q = aK + bL$ always exhibit constant returns to scale.

Example 2: Multiplicative Production Function**Example 2:** $Q = 0.5KL$

Again, we increase both K and L by m and create a new production function.

$$Q' = 0.5(Km)(Lm) \quad (37)$$

$$= 0.5KLm^2 \quad (38)$$

$$= Qm^2 \quad (39)$$

Since $m > 1$, then $m^2 > m$. Our new production has increased by more than m , so we have **increasing returns to scale**.

If $m = 2$:

$$m^2 = 4 > 2$$

Output quadruples when inputs double!

When the sum of exponents exceeds 1, we get increasing returns to scale. Here: $1 + 1 = 2 > 1$

Example 3: Cobb-Douglas with Decreasing Returns**Example 3:** $Q = K^{0.3}L^{0.2}$

Again, we increase both K and L by m and create a new production function.

$$Q' = (Km)^{0.3}(Lm)^{0.2} \quad (40)$$

$$= K^{0.3}L^{0.2}m^{0.5} \quad (41)$$

$$= Qm^{0.5} \quad (42)$$

Because $m > 1$, then $m^{0.5} < m$, our new production has increased by less than m , so we have **decreasing returns to scale**.

If $m = 2$:

$$m^{0.5} = \sqrt{2} \approx 1.414 < 2$$

Output increases by only 41.4% when inputs double.